

Model Evaluation Report

Project Name: Gneh Detection System

Date: October 2023

Model Version: YOLOv8n (Nano)

1. Executive Summary

This report presents the evaluation of a custom object detection model developed to distinguish between Gneh and NosGneh. The model is based on the YOLOv8 architecture and was trained using a labeled dataset hosted on Roboflow. The final evaluation results demonstrate excellent performance in terms of accuracy, precision, and recall, confirming the model's suitability for deployment in a real-time Streamlit web application.

2. Training Environment & Parameters

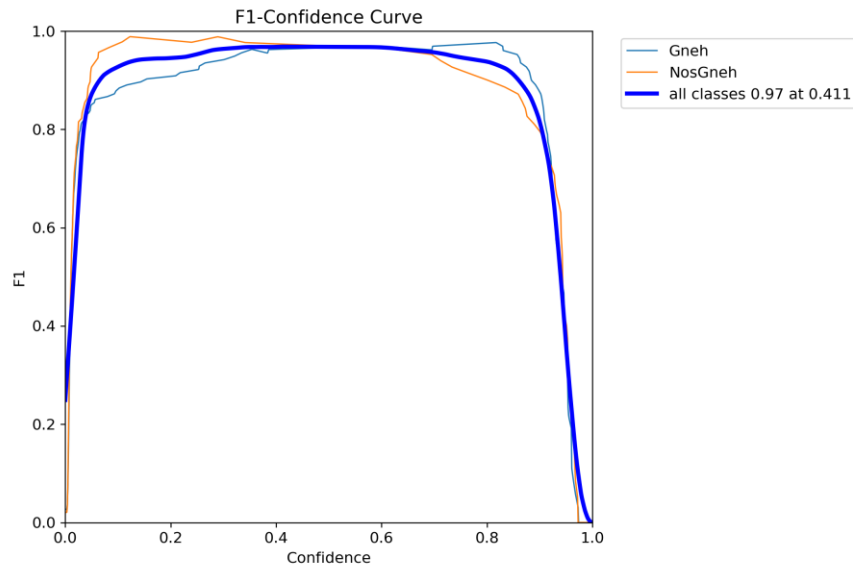
The model was trained using Google Colab with the following setup:

- Hardware: NVIDIA T4 GPU
- Base Model: yolov8n.pt (pre-trained on COCO dataset)
- Number of Epochs: 30
- Image Resolution: 640 × 640 pixels
- Optimization Strategy: Automatic optimizer selection (SGD / Adam)

3. Performance Metrics

The final evaluation metrics recorded at the end of the training process are as follows:

- mAP@50: 0.995 (Excellent detection accuracy at 50% IoU threshold)
- mAP@50–95: 0.824 (High performance across stricter IoU thresholds)
- Precision: 0.993 (Very low false-positive rate)
- Recall: 1.000 (All target objects were successfully detected)



4. Loss Analysis

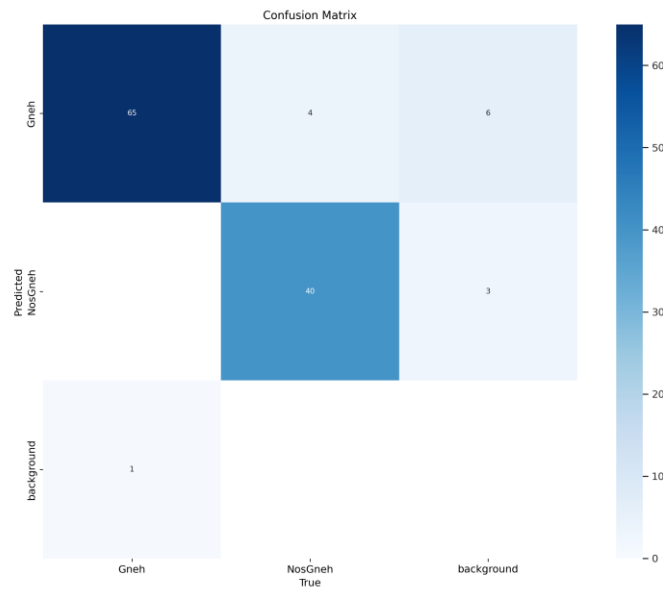
- Box Loss: Demonstrated a steady decline, indicating improved bounding box localization.
- Classification Loss: Reached near-zero values early in training, confirming the strong visual distinction between Gneh and NosGneh.
- Distribution Focal Loss (DFL): Gradual reduction throughout training, enhancing bounding box boundary precision.

5. Visual Evaluation (Inference)

Qualitative analysis during validation revealed:

- High confidence scores, typically exceeding 0.90.

- Tight and accurate bounding box placement with minimal localization error.
- Successful detection on unseen validation images, indicating strong generalization capability.



6. Conclusion

The YOLOv8n model has successfully converged and meets all predefined performance requirements. The model was exported to ONNX format, resulting in a lightweight file size of approximately 11.7 MB. This optimized format enables efficient real-time inference and seamless integration within a Streamlit-based deployment environment.